

Markscheme

Mathematics: analysis and approaches
Practice question bank

135. (a)

Given that $|z_1| = 6$ and $\arg(z_1) = \frac{\pi}{3}$, and $z_2 = 2 - 2i$.

Find $|z_1 z_2|$ and $\arg(z_1 z_2)$, giving the argument in radians correct to 3 significant figures.

$$|z_2| = \sqrt{8} \approx 2.828, \arg(z_2) = -\frac{\pi}{4} \quad (M1) \text{ A1}$$

$$|z_1 z_2| = |z_1| |z_2| = 6 \times 2.828 \quad (M1)$$

$$\approx 16.97 \quad \text{A1}$$

$$\arg(z_1 z_2) = \frac{\pi}{3} - \frac{\pi}{4} = \frac{\pi}{12} \approx 0.262 \quad \text{A1}$$

[6 marks]